

One Page Recommendation

Purpose

The model is designed to help the claims team identify which insurance claims are worth investigating for possible fraud. Fraud is relatively rare in the current data: 295 of 8,000 claims, or approximately 3.69%, are fraudulent.

A simple system that treated every claim as legitimate would appear to achieve 96.31% accuracy but would identify no fraudulent claims. For this reason, the model was evaluated using measures focused on finding rare fraud cases rather than accuracy.

Model performance

I compared a Decision Tree, Random Forest and XGBoost using the same features, preprocessing and train/validation split. The Random Forest produced the strongest validation PR-AUC in this experiment, approximately 0.166, compared with approximately 0.085 for the Decision Tree and 0.127 for XGBoost.

After the model-selection decision was made, the Random Forest was retrained on the development data and evaluated once on the untouched test set. Its test PR-AUC was approximately 0.142, compared with a random-ranking baseline of approximately 0.037. This indicates that the model provides useful ranking information above random selection.

Recommended operating point

The business can investigate approximately 200 claims per month. A threshold chosen only from financial costs would flag too many claims for the investigation team.

Using the 20:1 relative cost of missing fraud compared with investigating a legitimate claim, the cost-only threshold was approximately 0.062. However, that threshold would exceed the investigation capacity substantially.

I therefore recommend an operating threshold of approximately 0.20, based on the capacity-constrained validation threshold. This corresponds to reviewing roughly the highest-risk 2.5% of claims, or approximately 200 claims per 8,000 claims. The exact production implementation should preferably use a daily or monthly top-risk quota so that the investigation queue cannot exceed its capacity.

What drives the predictions

The exploratory analysis found higher observed fraud rates among higher-value claims, claims with more previous claims, claims reported after longer delays, and claims submitted through the Broker channel. SHAP analysis is used to determine whether these same variables are important to the trained model and to explain individual flagged claims.

The model should therefore not be treated as simply saying that a claim is "fraudulent." Instead, it identifies claims whose observed characteristics resemble patterns associated with fraud in the training data. Investigators should use the explanations as supporting evidence rather than as proof of fraud.

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Limitations

First, the fraud rate is only 3.69%, so the model operates in a highly imbalanced environment. Even with useful ranking performance, many flagged claims will still be legitimate. The investigation team therefore needs a controlled queue rather than treating every flagged claim as confirmed fraud.

Second, the model's probability calibration requires further validation before the score is used directly in a numerical formula such as risk score multiplied by claim amount. Calibration did not improve the Brier score in the current validation experiment, so I would not yet treat the raw probability as a guaranteed real-world probability.

Next week

With another week, I would test the model on a time-separated future period to check whether its performance remains stable. I would also investigate calibration using a larger representative calibration sample, monitor performance at the 200-claim capacity point, and review the highest-impact SHAP features with fraud investigators.

The next production test should therefore focus not only on overall model metrics, but also on how many genuine fraud cases are found among the approximately 200 claims the team can realistically investigate.